

PRAGATI LAB INFORMATION SYSTEM

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ABSTRACT

The Pragati Lab Information System is a simple C++-based console application designed to provide quick access to laboratory details in an academic environment. The system allows users to input a floor and room number to retrieve information such as the number of systems, fans, lights, chairs, projectors, and air conditioners available in a lab. This reduces dependency on manual inquiries and improves efficiency in accessing lab-related data.

Introduction

In many institutions, students and staff often need information about laboratories such as infrastructure and availability. Traditionally, this requires asking lab assistants or checking manually, which is time-consuming.

This project aims to solve this problem by:

- Providing instant lab details

- Reducing manual effort

- Improving accessibility of lab data

The system is implemented using C++, making it lightweight and easy to run on any system.

Flowchart (Text Representation)

START



Input Floor



Input Room Number



Check Floor & Room Match?



YES



NO



Display
Details



Show Error
Message



END

ALGORITHM

- 1.Start
- 2.Enter username and password
- 3.If login is correct → show menu
- 4.Else → show "Invalid Login" and go back
- 5.Show menu options
 - Repeat until user selects Logout:
 - If Student Management → manage students
 - If Lab Management → manage labs
 - If Attendance → mark attendance
 - If System Status → view systems
 - If Reports → generate reports
- 6.Stop

Methodology

The project follows a simple procedural approach:

The user provides input through the console

The system processes input using conditional statements (if-else)

Based on matching conditions, relevant lab data is displayed

If no match is found, an error message is shown

This approach ensures:

Simplicity

Fast execution

Easy modification and scalability

RESULTS

```
main.c... Output [Icons] [Run]
Enter Floor (ground / first / third): first
Enter Room Number / Lab Code: F18-19

Marvin Minsky Lab:
Systems: 72
Chairs: 75
Fans: 17
AC: 4
Lights: 9
Projector: 1

=== Code Execution Successful ===
```

```
main.c... Output [Icons] [Run]
Enter Floor (ground / first / third): third
Enter Room Number / Lab Code: T-2

Engineering Drawing Lab:
Systems: 66
Fans: 6
Chairs: 68
Lights: 6
Projector: 1
AC: 2

=== Code Execution Successful ===
```

```
main.c... Output [Icons] [Run]
Enter Floor (ground / first / third): first
Enter Room Number / Lab Code: F-9

Invalid Floor or Room!

=== Code Execution Successful ===
```

Conclusion

The Pragati Lab Information System successfully provides an efficient way to access lab details. It simplifies the process of retrieving information and minimizes manual effort. The project demonstrates the practical use of C++ in solving real-world problems in an academic environment.

Future Work

This project can be improved by:

- Adding a Graphical User Interface (GUI)
- Storing data in a database
- Developing a web or mobile application
- Adding search and filter options
- Including real-time lab availability status